WEEK 8: DATABASE RELATIONSHIP ANALYSIS USING JOINS

1. AIM

To analyze relationships between Customer, Product, Orders, Order_Details, and Payment tables using SQL joins and generate business reports from the E-Commerce Database.

2. IMPLEMENT INNER JOIN

INNER JOIN: Returns only matching records from both tables.

2.1 Display Customer Details Along with Their Orders

```
SQL> SELECT *
  2  FROM Customer C
  3  INNER JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            EMAIL          PHONE CREATED_A   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT ORDER_
STATUS
-------------------- -------------------- -------------------- ------------------------------ ---------- ---------- ---------- -------------------- ----------
 ------------ --------------------
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1001 C-10001              20-SEP-26
 5000 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1002 C-10001              21-SEP-26
 3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1003 C-10001              22-SEP-26
 8400 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1004 C-10001              23-SEP-26
18000 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1005 C-10001              24-SEP-26
 4500 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1006 C-10001              25-SEP-26
 2400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1007 C-10001              25-SEP-26
 3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1008 C-10001              26-SEP-26
 5600 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1009 C-10001              26-SEP-26
 5400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1010 C-10001              26-SEP-26
 3600 Delivered

10 rows selected.
```

Explanation: Displays customer details along with their corresponding order information.

2.2 Display Order Details with Payment Information

```
SQL> SELECT *
  2  FROM Orders O
  3  INNER JOIN Payment P
  4  ON O.Order_ID = P.Order_ID;

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
PAYMENT_ID       ORDER_ID PAYMENT_D PAYMENT_AMOUNT PAYMENT_METHOD       PAYM
ENT_STATUS
---------- -------------------- --------- ------------ --------------------
---------- ---------- --------- ------------- -------------------- --------
------------
      1001 C-10001              20-SEP-26         5000 Pending
    1         1001 01-OCT-26           5000 UPI                  Success
      1002 C-10001              21-SEP-26         3500 Shipped
    2         1002 02-OCT-26           3500 Card                 Pending
      1003 C-10001              22-SEP-26         8400 Delivered
    3         1003 03-OCT-26           8400 Net Banking          Failed
      1004 C-10001              23-SEP-26        18000 Pending
    4         1004 04-OCT-26          18000 UPI                  Success
      1005 C-10001              24-SEP-26         4500 Delivered
    5         1005 05-OCT-26           5000 Cash                 Pending
      1006 C-10001              25-SEP-26         2400 Pending
    6         1006 06-OCT-26           7200 Card                 Success
      1007 C-10001              25-SEP-26         3500 Shipped
    7         1007 07-OCT-26           6200 Net Banking          Success
      1008 C-10001              26-SEP-26         5600 Delivered
    8         1008 08-OCT-26           9500 UPI                  Pending
      1009 C-10001              26-SEP-26         5400 Pending
    9         1009 09-OCT-26          11000 Card                 Success

9 rows selected.
```

Explanation: Displays order details along with the corresponding payment information.

2.3 Retrieve Products Purchased by Customers

```
SQL> SELECT C.Customer_ID, P.Product_Name
  2  FROM Customer C
  3  INNER JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID
  5  INNER JOIN ORDER_DETAILS D
  6  ON O.Order_ID = D.Order_ID
  7  INNER JOIN Product P
  8  ON D.Product_ID = P.Product_ID;

CUSTOMER_ID          PRODUCT_NAME
-------------------- ------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------
C-10001              Gaming Headset
C-10001              Mechanical Keyboard
C-10001              Gaming Controller
C-10001              Gaming Monitor
C-10001              Bluetooth Speaker
C-10001              Wireless Mouse
C-10001              Mechanical Keyboard
C-10001              Gaming Controller
C-10001              Gaming Controller
C-10001              Power Bank

10 rows selected.
```

Explanation: Displays customer IDs and the names of products purchased by them.

3. IMPLEMENT LEFT JOIN

LEFT JOIN: Returns all records from the left table and matching records from the right table.

3.1 Display All Customers Including Customers Who Have Not Placed Orders

```
SQL> SELECT *
  2  FROM Customer C
  3  LEFT JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            EMAIL          PHONE CREATED_A   ORDER_ID CUSTOMER_ID           ORDER_DAT TOTAL_AMOUNT ORDER_
STATUS
-------------------- -------------------- -------------------- ------------------------------ ---------- --------- ---------- -------------------- ---------
 ------------ --------------------
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1001 C-10001              20-SEP-26
 5000 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1002 C-10001              21-SEP-26
 3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1003 C-10001              22-SEP-26
 8400 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1004 C-10001              23-SEP-26
18000 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1005 C-10001              24-SEP-26
 4500 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1006 C-10001              25-SEP-26
 2400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1007 C-10001              25-SEP-26
 3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1008 C-10001              26-SEP-26
 5600 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1009 C-10001              26-SEP-26
 5400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com          9876543210 14-FEB-23       1010 C-10001              26-SEP-26
 3600 Delivered

10 rows selected.
```

Explanation: Displays customers who have not placed any orders.

Output: No rows selected.

3.3 Display All Products Including Products with No Sales

```
SQL> SELECT P.*
  2  FROM Product P
  3  LEFT JOIN ORDER_DETAILS D
  4  ON P.Product_ID = D.Product_ID
  5  WHERE D.Product_ID IS NULL;

PRODUCT PRODUCT_NAME
      CATEGOR       PRICE STOCK_QUANTITY
------- ------------------------------------------------------------------------
--------------------------------------------------------------------------------
------ ------- ---------- --------------
P100005 Smartphone
      C103          25000              8
```

Explanation: Displays products that do not have matching records in the order details table.

4. IMPLEMENT RIGHT JOIN

RIGHT JOIN: Returns all records from the right table and matching records from the left table.

4.1 Display All Orders with Customer Information

```
SQL> SELECT *
  2  FROM Customer C
  3  RIGHT JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID          FIRST_NAME           LAST_NAME            EMAIL          PHONE CR
ID           ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
-------------------- -------------------- -------------------- ----------------------
 ---------- -------------------- --------- ------------ --------------------
C-10001              Ricky                Ponting              ricky200@edss.com
   1010 C-10001              26-SEP-26     3600 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com
   1009 C-10001              26-SEP-26     5400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com
   1008 C-10001              26-SEP-26     5600 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com
   1007 C-10001              25-SEP-26     3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com
   1006 C-10001              25-SEP-26     2400 Pending
C-10001              Ricky                Ponting              ricky200@edss.com
   1005 C-10001              24-SEP-26     4500 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com
   1004 C-10001              23-SEP-26    18000 Pending
C-10001              Ricky                Ponting              ricky200@edss.com
   1003 C-10001              22-SEP-26     8400 Delivered
C-10001              Ricky                Ponting              ricky200@edss.com
   1002 C-10001              21-SEP-26     3500 Shipped
C-10001              Ricky                Ponting              ricky200@edss.com
   1001 C-10001              20-SEP-26     5000 Pending

10 rows selected.
```

Explanation: Displays all orders along with matching customer information.

4.2 Find Orders Where Customer Details Are Missing

```
SQL> SELECT O.*
  2  FROM Customer C
  3  RIGHT JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID
  5  WHERE C.Customer_ID IS NULL;

no rows selected
```

Explanation: Displays orders that do not have matching customer records.

Output: No rows selected.

4.3 Display All Payment Records with Order Details

```
SQL> SELECT *
  2  FROM Orders O
  3  RIGHT JOIN Payment P
  4  ON O.Order_ID = P.Order_ID;

  ORDER_ID CUSTOMER_ID          ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT_ID
T_AMOUNT PAYMENT_METHOD PAYMENT_STATUS
---------- -------------------- --------- ------------ -------------------- ----------
--------- -------------------- --------------------
      1001 C-10001              20-SEP-26         5000 Pending                       1
    5000 UPI                    Success
      1002 C-10001              21-SEP-26         3500 Shipped                       2
    3500 Card           Pending
      1003 C-10001              22-SEP-26         8400 Delivered                     3
    8400 Net Banking            Failed
      1004 C-10001              23-SEP-26        18000 Pending                       4
   18000 UPI                    Success
      1005 C-10001              24-SEP-26         4500 Delivered                     5
    5000 Cash           Pending
      1006 C-10001              25-SEP-26         2400 Pending                       6
    7200 Card           Success
      1007 C-10001              25-SEP-26         3500 Shipped                       7
    6200 Net Banking            Success
      1008 C-10001              26-SEP-26         5600 Delivered                     8
    9500 UPI                    Pending
      1009 C-10001              26-SEP-26         5400 Pending                       9
   11000 Card           Success

9 rows selected.
```

Explanation: Displays all payment records along with matching order details.

5. RETRIEVE COMPLETE ORDER DETAILS

Task: Generate a complete order report containing customer name, product name, quantity, order date, total amount, and payment status.

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         P.Product_Name,
  3         D.Quantity,
  4         O.Order_Date,
  5         O.Total_Amount,
  6         Pay.Payment_Status
  7  FROM Customer C
  8  JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID
 10  JOIN ORDER_DETAILS D
 11  ON O.Order_ID = D.Order_ID
 12  JOIN Product P
 13  ON D.Product_ID = P.Product_ID
 14  JOIN Payment Pay
 15  ON O.Order_ID = Pay.Order_ID;

CUSTOMER_NAME                          PRODUCT_NAME
-------------------------------------- ----------------------------------------
----------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Ricky Ponting                          Gaming Headset
         2 20-SEP-26         5000 Success

Ricky Ponting                          Wireless Mouse
         2 25-SEP-26         2400 Success

Ricky Ponting                          Mechanical Keyboard
         1 25-SEP-26         3500 Success


CUSTOMER_NAME                          PRODUCT_NAME
-------------------------------------- ----------------------------------------
----------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Ricky Ponting                          Mechanical Keyboard
         1 21-SEP-26         3500 Pending

Ricky Ponting                          Gaming Controller
         1 26-SEP-26         5400 Success

Ricky Ponting                          Gaming Controller
         2 26-SEP-26         5600 Pending


CUSTOMER_NAME                          PRODUCT_NAME
-------------------------------------- ----------------------------------------
----------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Ricky Ponting                          Gaming Controller
         3 22-SEP-26         8400 Failed

Ricky Ponting                          Bluetooth Speaker
         1 24-SEP-26         4500 Pending

Ricky Ponting                          Gaming Monitor
         1 23-SEP-26        18000 Success


9 rows selected.
```

Explanation: Combines customer, order, product, order details, and payment information into a complete order report.

6. DISPLAY CUSTOMER PURCHASE HISTORY

6.1 Display All Products Purchased by Customers

```
SQL> SELECT C.Customer_ID, P.Product_Name
  2  FROM Customer C
  3  JOIN Orders O
  4  ON C.Customer_ID = O.Customer_ID
  5  JOIN ORDER_DETAILS D
  6  ON O.Order_ID = D.Order_ID
  7  JOIN Product P
  8  ON D.Product_ID = P.Product_ID;

CUSTOMER_ID          PRODUCT_NAME
-------------------- ----------------------------------------------------------------------------------------------------
C-10001              Gaming Headset
C-10001              Mechanical Keyboard
C-10001              Gaming Controller
C-10001              Gaming Monitor
C-10001              Bluetooth Speaker
C-10001              Wireless Mouse
C-10001              Mechanical Keyboard
C-10001              Gaming Controller
C-10001              Gaming Controller
C-10001              Power Bank

10 rows selected.
```

Explanation: Displays the products purchased by each customer.

6.2 Calculate Total Amount Spent by Each Customer

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         SUM(O.Total_Amount) AS Total_Spent
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
C-10001              Ricky                      59900
```

Explanation: Counts the number of orders placed by each customer.

Display Latest Purchase Details of Customers

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         O.Total_Amount
  6  FROM Customer C
  7  JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  WHERE O.Order_Date = (
 10      SELECT MAX(O2.Order_Date)
 11      FROM Orders O2
 12      WHERE O2.Customer_ID = C.Customer_ID
 13  );

CUSTOMER_ID          FIRST_NAME           ORDER_ID ORDER_DAT TOTAL_AMOUNT
-------------------- -------------------- ---------- --------- ------------
C-10001              Ricky                      1010 26-SEP-26         3600
C-10001              Ricky                      1008 26-SEP-26         5600
C-10001              Ricky                      1009 26-SEP-26         5400
```

Explanation: Displays the latest order details for each customer.

7. GENERATE MULTI-TABLE BUSINESS REPORTS

Report 1: Customer Order Report

Required details: Customer name, Order ID, Order date, and Order status.

```
SQL> SELECT C.First_Name || ' ' || C.Last_Name AS Customer_Name,
  2         O.Order_ID,
  3         O.Order_Date,
  4         O.Order_Status
  5  FROM Customer C
  6  JOIN Orders O
  7  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_NAME                            ORDER_ID ORDER_DAT ORDER_STATUS
---------------------------------------- ---------- --------- --------------------
Ricky Ponting                                  1001 20-SEP-26 Pending
Ricky Ponting                                  1002 21-SEP-26 Shipped
Ricky Ponting                                  1003 22-SEP-26 Delivered
Ricky Ponting                                  1004 23-SEP-26 Pending
Ricky Ponting                                  1005 24-SEP-26 Delivered
Ricky Ponting                                  1006 25-SEP-26 Pending
Ricky Ponting                                  1007 25-SEP-26 Shipped
Ricky Ponting                                  1008 26-SEP-26 Delivered
Ricky Ponting                                  1009 26-SEP-26 Pending
Ricky Ponting                                  1010 26-SEP-26 Delivered

10 rows selected.
```

Explanation: Displays customer names along with their order IDs, dates, and statuses.

Report 2: Sales Report

Required details: Product name, quantity sold, and total revenue.

```
SQL> SELECT P.Product_Name,
  2         SUM(D.Quantity) AS Quantity_Sold,
  3         SUM(D.Quantity * P.Price) AS Total_Revenue
  4  FROM Product P
  5  JOIN ORDER_DETAILS D
  6  ON P.Product_ID = D.Product_ID
  7  GROUP BY P.Product_Name;

PRODUCT_NAME
      QUANTITY_SOLD TOTAL_REVENUE
--------------------------------------------------------------------------------------------------------------------------------
---------------------------------- ------------- -------------
Gaming Headset
  2           5000
Gaming Controller
  6          16800
Power Bank
  2           3600
Mechanical Keyboard
  2           7000
Gaming Monitor
  1          18000
Bluetooth Speaker
  1           4500
Wireless Mouse
  2           2400

7 rows selected.
```

Explanation: Calculates the total quantity sold and the revenue for each product using product prices and quantities.

Report 3: Payment Analysis Report

Required details: Payment mode, number of transactions, and successful payments.

```
SQL> SELECT Payment_Method,
  2         COUNT(*) AS Total_Transactions,
  3         SUM(CASE
  4             WHEN Payment_Status = 'Successful' THEN 1
  5             ELSE 0
  6         END) AS Successful_Payments
  7  FROM Payment
  8  GROUP BY Payment_Method;

PAYMENT_METHOD       TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ------------------ -------------------
Net Banking                           2                   0
Card                                  3                   0
Cash                                  1                   0
UPI                                   3                   0
```

Explanation: Counts transactions and successful payments for each payment method.

Report 4: Customer Purchase Analysis

A. Find Top Purchasing Customers

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         SUM(O.Total_Amount) AS Total_Spent
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name
  8  ORDER BY Total_Spent DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
C-10001              Ricky                      59900
```

Explanation: Displays customers in descending order of their total spending.

B. Find Customers with Maximum Orders

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         COUNT(O.Order_ID) AS Total_Orders
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name
  8  ORDER BY Total_Orders DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
C-10001              Ricky                          10
```

Explanation: Displays customers in descending order of their number of orders.

C. Find Customers with Highest Spending

```
SQL> SELECT C.Customer_ID,
  2         C.First_Name,
  3         COUNT(O.Order_ID) AS Total_Orders
  4  FROM Customer C
  5  JOIN Orders O
  6  ON C.Customer_ID = O.Customer_ID
  7  GROUP BY C.Customer_ID, C.First_Name
  8  ORDER BY Total_Orders DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
C-10001              Ricky                          10
```

Explanation: Ranks customers according to their total spending, with the highest spending shown first.

8. RESULT

The SQL queries were executed using SQL*Plus on the E-Commerce Database. INNER JOIN, LEFT JOIN, and RIGHT JOIN were used to retrieve related information from multiple tables. Customer purchase history and multi-table business reports were generated.

Some queries **returned no rows because no matching records were available in the database.**

9. CONCLUSION

Thus, database relationship analysis was performed successfully using SQL joins. The practical demonstrated how to combine related tables, retrieve customer purchase information, analyze payment transactions, and generate business reports using SQL.